Analog-to-Digital Conversion (ADC) Lab

BME554L - Fall 2026 - Palmeri

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Learning Objectives

- Implement single-sample and buffered ADC data acquisitions.
- Implement fixed-reference and differential ADC inputs.
- Implement synchronous and asynchronous ADC acquisitions.

Git Best Practices

- Use best practices for version control (branching, commit messages, etc.).
- Do all development on a dedicated branch that is merged into `main` once it is functional.
- Commits should be very specific to the changes/additions you are making to your code. This will help you and others understand what you did and why you did it.

- On a given development branch, try to implement one small piece of functionality at a time, commit it, and then move on to the next piece of functionality.

> **Important**
>
> You do not want one, monolithic git commit right before you submit your project.

Teamwork

- You will work with your assigned lab partner for this lab.
- You will be expected to work together to complete the lab, using git and Gitlab to manage your code development and testing.
- Each person should add their lab partner as a `Developer` to their Gitlab repository so that they can push and pull code to the repository.
- Each person should be commiting and pushing code to their own repositories on their own development branches and use Merge Requests to merge their code into the `main` branch, which may be the only "common" branch between each person's repositories.

> **Tip**
>
> Consider using libraries instead of writing all code in `main.c` to make it easier for each person to work on different parts of the firmware and avoid merge conflicts.
> Remember to include your libraries in your `CMakeLists.txt` file so that they are compiled and linked into your firmware.

> **Warning**
>
> You cannot have all of the commits be made by one person and then say "but we worked together on it". Each person should be contributing to the code development and testing as represented with individual commits on their own branches, and this will be evaluated during grading.

Firmware Expectations

- All firmware should be written using the **State Machine Framework**.
- Choose your states for each part as a firmware engineer would, using what you have learned so far this semester.
- Timers, work queues, callbacks, and interrupts should be used as appropriate.
- All good coding practices developed this semester should be followed.
- Use logging to display state information and other relevant information, warnings, and errors. Debugging log messages can remain in the code, but the logging level should be submitted at the `INF` level.
- Include state and sequence diagrams that represent your firmware's functionality.

> **Note**
>
> You will continue to build upon the firmware from your previous labs, but you are replacing all functional specifications with those in this lab. If you want to use a "clean slate", you can start with a fresh `main.c` file.
> All states should be reconsidered given the new functional specifications.
> This significant change in functionality is represented with the version tags changing to a new major version number starting at `v2.x`.

ADC Single Sampling

Firmware Functional Specifications

- Do all development and testing for Part I on a development branch called `adc_single_sample`.
- `LED` and `BUTTON` number references below are based on Devicetree labels (not the annotation on the DK board itself).
- There should be a heartbeat LED (`LED0`) that blinks at 1 Hz with a 25% duty cycle at all times when the firmware is running.
- When `BUTTON0` is pressed, make a measurement using the `AIN0` channel of the ADC.
  - Use the `ADC_REF_INTERNAL` reference voltage.
  - Linearly map 0-3.0 V measured on the `AIN0` input to an `LED1` blink rate of 1-5 Hz (e.g., 0 V -> 1 Hz; 3.0 V -> 5 Hz) with a duty cycle of 10%.
  - `LED1` should remaining blinking for 5 seconds after the button has been pressed.

- `BUTTON0` should be disabled while the ADC measurement and `LED1` blinking.
- All errors should result in a transition to an `ERROR` state that can only be exited by hitting a reset button (`BUTTON3`).

Testing

For the following testing, you can use a power supply to provide known voltage input to `AIN0`.

- Quantify how linear the relationship is between the voltage applied to `AIN0` and the `LED1` blinking frequency by acquiring data to perform a linear regression of these variables and analyzing the resultant R^2, slope and intercept.

> **Tip**
>
> Remember that single data points for any single input voltage pair is not adequate; multiple measurements should be made, sweeping increasing and decreasing voltages to evaluate hysteresis, and error bars presented on all plots.

> **Tip**
>
> The `bme554.py` library has a functions called `plot_with_fit()` and `calculate_confidence_intervals()` that can help.

- Quantify the accuracy of the `LED1` 10% duty cycle for each frequency that you measure.
- Quantify the accuracy of `LED1` 5 second ontime duration.
- Present all of these data, your analysis and your interpretation in a new Jupyter notebook named `testing/testing_adc_single_sample.ipynb`.

Commit-n-Merge Single Sample Acquisition

- Update your state and sequence diagrams.
- Once your firmware development and testing are done, merge your completed `adc_single_sample` branch into your `main` branch using a Merge Request on Gitlab.
- Create an annotated tag (`v2.0.0`) on your `main` branch of this merged branch.
- Create an Issue assigned to Dr. Palmeri indicating that this tag is ready for review.

Buffered Differential ADC Sampling

Firmware Functional Specifications

> **Tip**
>
> If you are working on this part before the previous part is complete, you can use the `adc_single_sample` branch as your starting point for this branch.

In a new development branch called `diff_adc`, add the following functionality to your firmware:

- When you press `BUTTON1`, add an additional differential ADC measurement using `AIN1` and `AIN2`.
- Choose the reference voltage, gain, bit depth, acquisition time and sampling interval to adequately sample at least 20 cycles of a 10 Hz sinusoidal signal ($V_{pp} = 2$ V).

> **Tip**
>
> Be sure to add a DC offset of `VDD`/2 to your input signal so that the signal is within the common-mode voltage range of the ADC.

> **Tip**
>
> Have your sampling interval be configured using a macro so that you can easily adjust it for testing purposes.
> Choose your ideal sampling interval based on the fact that you are only sampling a finite number of cycles (i.e., Nyquist frequency is not the only consideration) but without clobbering your SRAM with an unnecessarily large buffer.

- Implement the `extra_sampling` buffering of the ADC for this differential sinusoidal signal measurement so that all data are stored in an array in a single `adc_read_dt()` call (i.e., you are not using a kernel call for every sample).
- Disable the `BUTTON1` while your device is reading `AIN1` and `AIN2` and performing calculations below.
- Write a **library** called `calc_freq.[h/c]` that calculates the frequency of the sinusoidal signal in your buffer, ignoring the spectral content associated with the finite temporal window. Have this frequency displayed using `LOG_INF()` in the console.
- Write log messages to your serial terminal that displays a HEX array of the buffered ADC samples.

- Turn on `LED2` for 1 second after the buffered ADC sequence is complete.
- Update your state and sequence diagrams for this new functionality.

> **Warning**
>
> Be careful for the log message memory consumption! You may need to adjust the stack size of the appopriate thread if you are getting memory-related firmware faults.

> **Note**
>
> Remember the data will be saved using two's complement.

Testing

Nominal Sampling Interval

- Input a 10 Hz sinusoidal signal with a 2 V V_{pp} into the differential `AIN1` and `AIN2` inputs.
- Create a plot of your input signal and the buffered ADC samples (using the HEX array output).

> **Tip**
>
> There are helper functions in `bme554.py` called `read_hex_data()` and `unwrap_twos_complement()`.
> The HEX data array from the terminal can be saved to a text file in your git repository to read into your Jupyter notebook for analysis. Remember to not use absolute paths to your data file in your Jupyter notebook so that your lab partner (and teaching team) can run the notebook on their computer without having to change the path.

- Discuss any differences between the input signal and your sampled signal.
- Calculate the frequency of your sampled signal and compare it to your input frequency.
- Add this analysis to a new Jupyter notebook named `testing/testing_adc_sync.ipynb`.

> **Warning**
>
> Merging Jupyter notebooks from different branches can be tricky and usually requires some extra tools or manual intervention. We are creating separate Jupyter notebooks for each part of this lab in case you jump to code development in a different part before finishing the testing for the previous part.

Aliased Sampling

- Reduce your sampling frequency to 5 Hz and repeat the Nominal Sampling Interval measurements and analysis.
- Comment on what you see in your sampled signal, what the resultant frequency looks like and why your signal looks so different than the input signal.
- Discuss the "ideal" sampling frequency relative to your input frequency for your device.

Commit-n-Merge Buffered Acquisition

- Merge your completed `diff_adc` branch into your `main` branch using a Merge Request on Gitlab.
- Create an annotated tag of your `main` branch with all part of this lab merged in called `v2.1.0`.
- Create an Issue assigned to Dr. Palmeri indicating that this tag is ready for review.

Asynchronous ADC Sampling

Firmware Functional Specifications

> **!** Important
>
> Make sure that you set your sampling interval back to the nominal value, not the value you set for demonstrating sampling aliasing.

In a new development branch called `async_adc`, add the following functionality to your firmware:

- Replicate the functionality of your blocking, synchronous buffered ADC sequence acquisition now using an **asynchronous** acquisition that populates the same array that was populated with the synchronous approach.
- Use `AIN3` and `AIN4` for the differential ADC input that will be read using this asynchronous approach.
- Use `BUTTON2` to trigger the asynchronous ADC sequence acquisition and illuminate `LED3` for 1 second after the sequence is complete.
- Utilize the Zephyr ADC async enumerations presented in lecture to modulate the behaviour of the asynchronous callback function.

- Detect the raised kernel poll (`k_poll`) signal to indicate when the ADC sequence is complete.
- Implement a timeout mechanism to handle the case where the ADC sequence does not complete within a reasonable time frame.

Testing

- Compare your sampled signals using your asynchronous sampling approach to your buffered approach in a new Jupyter notebook named `testing/testing_adc_async.ipynb`.
- Discuss the benefits and drawbacks of the asynchronous ADC sequence sampling compared to the synchronous sampling.

Commit-n-Merge Asynchronous Sampling

- Merge your completed `async_adc` branch into your `main` branch using a Merge Request on Gitlab.
- Create an annotated tag of your `main` branch with all part of this lab merged in called `v2.2.0`.
- Create an Issue assigned to Dr. Palmeri indicating that this tag is ready for review.

How to Ask for Help

1. If you have a general / non-coding question, you should ask your TAs / Dr. Palmeri on Ed to allow any of them to respond in a timely manner.
2. Push you code to your GitLab repository, ideally with your active development on a non-`main` branch.
3. Create an Issue in your repository.
    - Add as much detail as possible as to your problem, and add links to specific lines / section of code when possible.
    - Assign the label "Bug" or "Question", as appropriate.
    - Be sure to specify what branch you are working on.
    - Assign the Issue to one of the TAs.
    - If your TA cannot solve your Issue, they can escalate the Issue to Dr. Palmeri.
4. You will get a response to your Issue, and maybe a new branch of code will be pushed to help you with some example syntax that you can use `git diff` to visualize.